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Clinical Application of Large Language Models for Intervention Plan Development in Speech-Language Pathology

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ABSTRACT

Purpose: This study investigates the speech and language intervention plan outputs generated by six different artificial intelligence (AI) tools powered by large language models (LLMs), currently available for clinical writing in the field of speech-language pathology. This study aims to evaluate the potential applications and limitations of these AI tools, as well as their ability to provide relevant and reliable information for developing intervention plans.
Method: Using a mixed design including both quantitative and qualitative analyses, this study compared the performance outputs of the six AI tools across three fictional clinical cases, each involving different types of speech and language disorders in 5-year-old children. Two types of command prompts, each with three levels of input specificity, were used to generate AI outputs.
Results: Results revealed that the intervention plans generated by these AI tools were rated between *Needs Improvement* and *Meets Expectations* in terms of clinical knowledge and competency. Detailed and structured command prompts than general prompts yielded outputs with higher ratings, while the specificity of case information did not consistently influence the outputs. Each AI tool demonstrated unique strengths and limitations in supporting the development of intervention plans.
Conclusion: The results of this study may serve as foundational data to provide insights into how clinicians, educators, and students in the field of speech-language pathology can appropriately and responsibly utilize existing AI resources when implementing these technologies into the development of intervention plans.

Since generative artificial intelligence (AI)—AI systems capable of generating content such as text, images, audio, and video (Sengar et al., 2024)—became widely accessible to the public in late 2022 with the emergence of AI tools like ChatGPT, there has been increasing interest and discussions about potential benefits, concerns, and future directions of AI usage in speech-language pathology and other related fields (Deka et al., 2024; Homolak, 2023; Price et al., 2024; Suh et al., 2024). Most popular AI tools today (e.g., ChatGPT) are powered by large language models (LLMs) like GPT-4 (Achiam et al., 2023). LLMs are deep learning models based on transformer architectures (Vaswani et al., 2017), trained on vast amounts of data to predict text sequences, enabling them to understand and generate human language effectively (Achiam et al., 2023). Throughout this article, "AI tools" will refer specifically to those built on top of LLMs, which offer user-friendly interfaces and some customization options.

With the growing potential for AI use in clinical settings, it is expected that clinicians will increasingly utilize AI tools for various purposes (Stokel-Walker & Van Noorden, 2023). Due to a reported significant shortage of speech-language pathologists (SLPs) and speech-language services for children with special needs in the U.S. public schools (Marante et al., 2023), the utilization of AI

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technologies may support both clinicians and clients through streamlining SLPs' documentation efforts, such as intervention planning and report writing, which in turn provides clinicians with additional time to directly work with their clients. In particular, the use of AI tools may be beneficial for intervention planning, which involves determining goals, selecting targets, identifying appropriate evidence-based approaches, designing sessions with materials and activities, and considering client-specific family or cultural-linguistic backgrounds.

On the other hand, there are growing concerns regarding the credibility and accountability of these LLM-based AI tools. While LLMs demonstrate remarkably strong performance, they inherently suffer from the problem of data hallucination—the generation of false, misleading, or nonsensical information that appears plausible but lacks a factual basis, stemming from the probabilistic nature of language modeling (Xu et al., 2024). Therefore, when using these AI tools to support intervention planning in the field of speech-language pathology, it is critical to understand the distinctive characteristics of the outputs generated by them. It is also essential to establish clearly defined parameters and guidelines for the scope and appropriate use of AI tools to procure reliable and pertinent outputs. Efforts have begun to develop specialized clinical AI systems in the field of speech-language pathology. *The ASHA Leader* recently featured discussions on the impact of AI in the field and its future implications (Law, 2023). The article introduced that multidisciplinary research teams comprising clinical practitioners, bioengineers, computer scientists, and data scientists have initiated projects aiming to develop specialized clinical AI tools based on extensive data sets of speech, language, hearing, and health information. These clinical AI tools may be able to provide significant advantages for assessment, diagnosis, and intervention for speech-language pathology and other related fields. However, the timeline for the availability and accessibility of reliable and professional clinical AI systems remains uncertain. Until the development of more specialized and easily accessible clinical AI tools is realized, it may be useful to explore currently available AI tools, understand their strengths and weaknesses, and discuss potential guidelines for the ethical and relevant use of these technologies.

Recent studies have reported on the applications of AI in the medical (Filienko et al., 2024; Liao et al., 2023; Pierce, 2023) or education fields (Park & Park, 2024). A recent systematic review of AI applications in clinical documentation has revealed that, while current AI tools offer promising targeted improvements, their widespread implementation as comprehensive documentation assistants may be hindered by high error rates. Researchers emphasized the critical need for rigorous evaluation of these tools and careful consideration of ethical and legal implications in health care settings, including speech-language pathology (Perkins et al., 2024). Lee et al. (2024), in their scoping review, highlighted the potential of AI in clinical documentation across health care disciplines, including speech-language pathology. While AI technologies demonstrate promise in enhancing efficiency, accuracy, and patient engagement, researchers emphasize the critical need for careful implementation strategies, continuous evaluation processes, and robust regulatory frameworks to address challenges and ensure patient safety in AI-assisted documentation (Lee et al., 2024).

Currently, research on the use of AI tools for clinical writing or intervention plan development in the field of speech-language pathology is highly limited, with only a few brief articles and tutorials addressing this topic (e.g., Price et al., 2024; Slavych et al., 2024). Price et al. (2024) evaluated ChatGPT-3.5's performance in developing treatment materials and found that, while the language activities created by the tool were useful, they were often inappropriate or insufficient for certain cases. Slavych et al. (2024), in their tutorial, discussed the use of ChatGPT to improve general writing skills for college students in communication sciences and disorders, rather than focusing on clinical writing. Both articles emphasized that clinicians and educators should use their expertise and clinical judgment in creating prompts when using AI tools, rather than relying solely on the tools.

Although some commercialized AI tools for therapy session planning in clinical settings have been available online, no studies have systematically analyzed whether currently available AI tools can provide accurate and reliable information for writing speech and language intervention plans. Additionally, the impact of prompt quality and detail on the output of clinical report writing is still unclear. The present study evaluates the intervention plan outputs generated by six different AI tools for various pediatric case scenarios involving speech and language difficulties. This study aims to identify the strengths and limitations of the currently available AI resources and support to enhance their informed application and ethical usage of these technologies in clinical practices.

The research questions are outlined as follows:

1. To what extent are currently available AI tools capable of generating effective speech-language therapy plans?
2. How do the characteristics and specificity of input prompts influence the content and relevance of speech and language intervention plans?
3. What are the strengths and limitations of speech and language intervention plan outputs across six different AI tools?

Method

This study compared the performance of outputs generated by six LLM-based AI tools across three fictional clinical cases. Five trained researchers assessed each AI-generated output using a grading rubric. The study was exempt from institutional review board approval as it did not involve human subjects or identifiable data.

AI Tool Selection

Six LLM-powered AI tools that can be potentially used for clinical report writing were selected for this study. These included four general-purpose AI tools and two specialized tools designed for therapy planning or activity generation. The selected tools were ChatGPT-3.5, ChatGPT-4 Omni (hereinafter, ChatGPT-4o), Copilot, Gemini, KiKi Creates by Ambiki, and MagicSchool.

Two versions of ChatGPT, developed by OpenAI, were included. ChatGPT-3.5 was chosen because it is one of the most widely used and freely accessible generative AI tools (OpenAI, 2023). ChatGPT-4o, OpenAI's latest and more advanced paid version, was selected for its enhanced performance compared to the free version (OpenAI, 2024). Both ChatGPT-3.5 and ChatGPT-4o share similar training data sources, but ChatGPT-4o introduces multimodal capabilities and includes data up to April 2023, whereas ChatGPT-3.5 has a cutoff date of January 2022 (OpenAI, 2024).

Copilot and Gemini were included because they show similar general-purpose AI models like ChatGPT but from different companies. Both models offer free and paid versions, but the free versions were used in this study. Copilot, developed by Microsoft, is designed to integrate with various Microsoft software and productivity tools (Spataro, 2023). Gemini, Google's AI model, generates outputs from various multimodal sources, leveraging Google's online resources for enhanced contextual understanding (Google DeepMind, 2023).

Researchers chose an AI-powered tool called KiKi Creates within Ambiki. Ambiki is a commercial clinical practice management platform collaboratively created by SLPs, physical therapists, and occupational therapists (Ambiki, n.d.). Of relevance to this research was the platform's key feature, KiKi Creates, an AI-powered tool designed to generate tailored content for therapy sessions and stimulus creation. KiKi Creates produces its outputs using input data by the users and user interactions, enabling therapists to rapidly adapt to children's specific interests or unexpected discussion topics during sessions. The specific training data for Ambiki and KiKi Creates are not publicly disclosed.

Lastly, researchers included MagicSchool, a free AI platform used in school settings, supporting teachers in developing lesson plans for kindergarteners through college students (MagicSchool, n.d.). Even though it was not a clinical writing tool directly, its generative lesson plan writing format and structure features may be relevant for clinical session plan writing. Because MagicSchool does not offer a single feature that provides output in a similar format to the other AI tools, researchers combined the outputs from three built-in features including Lesson Plan, Individualized Education Program Generator, and Accommodation Suggestion within MagicSchool to provide continuity across all output elements generated from the selected AI tools for the study. Table 1 presents a comprehensive comparison of the components of all six AI tools, delineating their key attributes, target demographics, pricing structures, and core functionalities.

Case Scenarios

Three fictional case scenarios of 5-year-olds with different types of speech and language disorders were developed by the researchers. Three case scenarios were (a) SSD: a child with functional speech sound disorders without any other functional or organic deficits; (b) LD: a child with language disorders as well as speech sound disorders, exhibiting limited expressive vocabulary and morphological

Table 1. Comparison of artificial intelligence tools selected for this study.

Tool	Primary purpose	Target users	Pricing models selected	Access URL
ChatGPT-3.5	General purpose	General public	Free version	chat.openai.com
ChatGPT-4o	General purpose	General public	Paid version	openai.com/plus
Copilot	General purpose, productivity assistance	General public	Free version	copilot.microsoft.com
Gemini	General purpose	General public	Free version	gemini.google.com
KiKi Creates by Ambiki	Specialized purpose (therapy practice management)	SLPs, OTs, PTs	Free trial	ambiki.com
MagicSchool	Specialized purpose (educational resource generation)	Educators	Free trial	magicschool.ai

Note. SLPs = speech-language pathologists; OTs = occupational therapists; PT = physical therapists.

problems; and (c) ASD: a child with autism spectrum disorders who demonstrated speech, language, and pragmatic difficulties. Each case included detailed background information, as well as standardized and nonstandardized assessment results. Full descriptions of an example SSD case with three different specificity levels can be found in Appendix A.

Input Prompts

A prompt in the context of LLMs refers to a textual input request that one provides to the tool to generate a response (Budiu et al., 2023). In this study, we designed a prompt composed of a command statement and a case description. The command statements used in this study were categorized into two types: *Simple Command* and *Framed Command*. Simple Commands involve general requests that do not require specific formatting. In contrast, Framed Commands involve more structured requests that require specific elements in the outputs, including therapy goals, evidence-based practice resources, therapy activities, client engagement, progress monitoring tools, and cultural and linguistic considerations. The command statements of the prompts used in this study are as follows:

- *Simple Command:* "Provide a speech-language therapy intervention plan for the following case."
- *Framed Command:* "Provide a detailed and individually customized speech-language therapy intervention plan, including 1) three therapy goals with specific targets, 2) evidence-based intervention strategies with 2 research references, 3) age-appropriate therapy activities suggestions and materials, 4) age-appropriate recommendations for client engagement, behavior and motivation suggestions, 5) progress monitoring tools for tracking goal progress for the following client case, and 6) culturally and linguistically responsive and inclusive practice recommendations."

These command statements were paired with a case description that varied across three levels of specific profile information: *Basic, Intermediate*, and *Detailed*. The *Basic* case descriptions included general information about the case. The *Intermediate* case descriptions provided specific problems and characteristics of the case's speech and language difficulties. The *Detailed* case descriptions contained comprehensive information including the case history, family history, and results from both standardized and nonstandardized tests. Based on the two types of command statements and the three levels of case descriptions, six different input prompts were defined as follows: Simple Command with Basic case description, Simple Command with Intermediate case description, Simple Command with Detailed case description, Framed Command with Basic case description, Framed Command with Intermediate case description, and Framed Command with Detailed case description. Six types of prompts were entered for three case scenarios (SSD, LD, and ASD) for each AI tool. Consequently, the researchers composed 18 prompts for each AI tool, resulting in 18 written intervention plans generated by each of the six AI tools examined in the study. Figure 1 outlines the structure of the prompts, which include two types of command statements and three levels of case descriptions for each disorder used in this study.

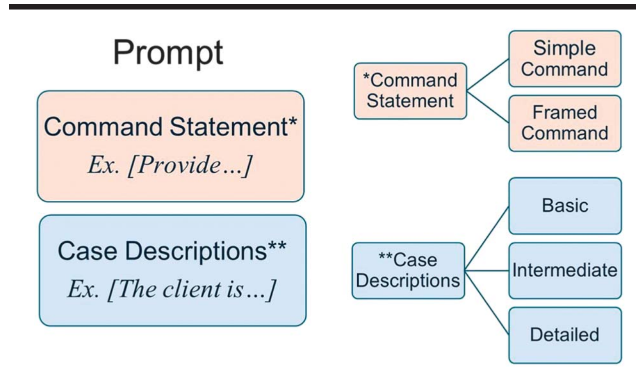
Generating Output Data

Researchers inputted 18 predefined prompts into each AI tool and generated intervention plans as outputs. Each output was saved into a document file for data analysis. To maintain consistent parameters across all six AI tools, no subsequent prompts nor interactions with the tools were conducted. Between each prompt input for each AI, previous inputs and outputs were deleted. If the AI tool provided an option to minimize or prevent self-training or internal learning from data history, researchers adjusted the settings accordingly. For example, both ChatGPT-3.5 and ChatGPT-4o offer options to customize user profiles to make the responses more targeted to user preferences and to disable internal learning from data history. Researchers selected these options to optimize output performances for speech-language pathology–related clinical writing and to prevent internal training from data history.

Grading Rubrics

The outputs generated by AI tools were evaluated using a grading rubric developed by the researchers. The grading rubric consisted of six objective items rated on a 5-point scale and one open-ended comment item. The six objective items evaluated the contents and clinical relevance

Figure 1. Prompt structure used in this study.



of therapy goals, evidence-based intervention strategies, therapy activities, client engagement, progress monitoring tools, and cultural, linguistic, and family considerations. This rubric was designed as a tool to rate the quality of the AI-generated intervention plans by paralleling them to the expected clinical knowledge and competency learning outcomes for intervention planning from graduate students in speech-language pathology. The rubric scaled from 1 to 5, where 5 = *Excellent*, 4 = *Good*, 3 = *Meets Expectations*, 2 = *Needs Improvements*, and 1 = *Unsatisfactory*. The open-ended question item asked researchers to describe, in one to two sentences, overall perceived characteristics, strengths, and limitations of each response from each AI tool. The grading rubric was attached in Appendix B.

Data Analysis

Five trained researchers independently rated the outputs from each AI tool using the grading rubric. Each researcher was blinded to the ratings given by other researchers until all the scoring was done. They entered their rating into an online scoring system, which exported all the data into Excel spreadsheets. Interrater reliability was evaluated using the intraclass correlation coefficient (ICC), which is appropriate for assessing interrater reliability among multiple raters with ordinal data (Koo & Li, 2016; Shrout & Fleiss, 1979). The ICC was calculated using a mean-rating (k = 5), absolute-agreement, and two-way random-effects model. The ICC was .927 (95% confidence interval [.760, .988]; p < .001), indicating excellent reliability among raters.

Descriptive statistics and nonparametric analysis were utilized to summarize the basic features of the data across the six AI tools. For quantitative analysis, average rating scores across researchers and their standard deviations were calculated for each objective-style item for each AI tool. Given that the data in this study comprised average scores from a small sample of raters (i.e., five researchers) across multiple levels of variables, statistical significance testing was not conducted. Instead, the results were reported descriptively. For qualitative analysis, each researcher independently identified the characteristics, strengths, and limitations of each AI tool and submitted their qualitative responses into an online survey form. After all researchers completed their qualitative feedback on each output, researchers analyzed the qualitative data using the open thematic coding method (Armstrong et al., 1997) to identify key themes and patterns. A list of the topics from the responses was compiled and then clustered into groups of similar topics by the researchers. After researchers completed the thematic coding for each item, they identified common descriptors that occurred three or more times in the analysis.

A total of 108 intervention plan output data were analyzed. These outputs were generated using six different prompts, each with varying levels of specificity, applied to cases of 5-year-old children with three different speech and language disorders (i.e., SSD, LD, and ASD) across six AI tools. The results included average rating scores and qualitative evaluations of the output performances.

Results

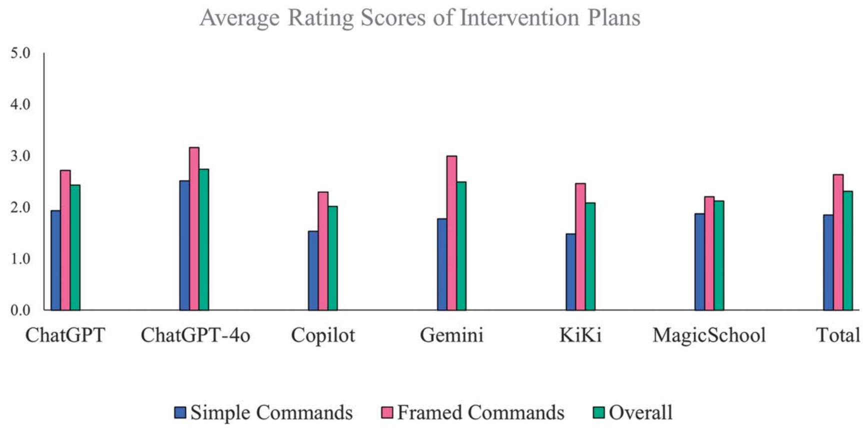
Average Rating Scores

This study analyzed the average rating scores of the intervention plan outputs generated by six AI tools. As described in the Data Analysis section, the grading rubric ratings were scaled from 1 to 5, with 5 = *Excellent*, 4 = *Good*, 3 = *Meets Expectations*, 2 = *Needs Improvement*, and 1 = *Unsatisfactory*. The overall average score from all six AI tools was 2.31 (SD = 0.55) indicating the intervention plan outputs generated by these AI tools were rated at a level between *Meets Expectations* and *Needs Improvement* in terms of clinical knowledge and competency required by speech-language pathology clinicians. Of six AI tools, ChatGPT-4o achieved the highest averaged score (M = 2.73, SD = 0.61) followed by Gemini (M = 2.49, SD = 0.59) and ChatGPT-3.5 (M = 2.43, SD = 0.58). MagicSchool (M = 2.12, SD = 0.52), KiKi Creates (M = 2.08, SD = 0.5), and Copilot (M = 2.01, SD = 0.52) scored lower average ratings compared to ChatGPT-4o, Gemini, and ChatGPT-3.5. These average scores were based on aggregated data across different types of disorders, prompts, and items in the plans. The results for each parameter are described in the following sections. The average rating scores of these six AI tools were displayed in Figure 2.

Rating Scores by Types of Disorders

All these AI tools, except Copilot and KiKi Creates, received slightly higher ratings in intervention plans for ASD than those for LD and SSD cases. For example, Gemini received an average score of 2.58 (SD = 0.66) for ASD, 2.47 (SD = 0.55) for LD, and 2.40 (SD = 0.56) for SSD. ChatGPT-3.5 received an average score of 2.56 (SD = 0.74) for ASD, 2.42 (SD = 0.55) for LD, and 2.31 (SD = 0.46) for SSD. However, when taking the standard deviations of the averaged scores into account, these small differences among average scores may not be significant nor meaningful. In ChatGPT-4o, which received the highest average rating, all three cases, including ASD (M = 2.75, SD = 0.64), LD (M = 2.74, SD = 0.54), and SSD (M = 2.72, SD = 0.65), yielded similar ratings. Overall, the results showed no substantial differences in the AI-

Figure 2. The average rating scores of artificial intelligence outputs by prompt types.



generated intervention plans across the three types of disorders. The summary table of the average scores and standard deviations for each disorder can be found in Table 2.

Rating Scores by Prompts

Rating Scores by Prompt Types

The average scores of intervention plan outputs generated by different prompts were analyzed. The results are summarized in Table 1. For all six AI tools, the outputs responding to prompts with *Framed* command statements (hereafter, Framed Command) received noticeably higher ratings (M = 2.66, SD = 0.56) than those from prompts with *Simple* command statements (hereafter, Simple Command; M = 1.93, SD = 0.54). Prompts with Framed Commands included command statements requiring a specified format in the outputs, whereas prompts with Simple Commands included general command statements only. Results showed that Simple Commands generated outputs at a level between *Unsatisfactory* and *Needs Improvement* while Frame Commands yielded outputs at a level between *Needs Improvement* and *Meet Expectations*.

This tendency was observed in all cases regardless of the types of disorders. For the outputs from prompts with Simple Commands, ChatGPT-4o received the highest score (M = 2.41, SD = 0.58) at a level between 2 and 3, indicating *Needs Improvement* and *Meet Expectations*. All other tools produced outputs below the *Needs Improvement* level. In contrast, for the outputs from prompts with Framed Commands, ChatGPT-4o (M = 3.06, SD = 0.64), Gemini (M = 2.97, SD = 0.59), and ChatGPT-3.5 (M = 2.82, SD = 0.60) received the highest average scores across cases. KiKi Creates (M = 2.61, SD = 0.55) produced outputs at a level between 2 and 3. Copilot (M = 2.31, SD = 0.49) and MagicSchool (M = 2.19, SD = 0.51) produced slightly lower rated outputs than other AI tools.

Among these six AI tools, Gemini and KiKi Creates exhibited the greatest difference in the average scores between Simple and Framed Commands. ChatGPT-4o and MagicSchool showed small differences in the average scores between outputs for Simple and Framed Commands. When using prompts with Simple Commands, ChatGPT-4o generated outputs with the highest ratings for all disorder cases. For prompts with Frame Commands, Gemini received the highest score for ASD cases, while ChatGPT-4o received the highest score for LD cases. The average scores for the outputs from Simple Commands were around 1.87, while those from Framed Commands were around 2.64. However, it should be noted that the highest ratings of all AI tools for Simple Command prompts across all three cases were still at a level of 2 (*Needs Improvement*), while the scores for Framed Command prompts reached at a level of 3 (*Meet Expectations*). These results indicate that using prompts with Framed Commands may be more likely to obtain relevant intervention plans for most AI models analyzed in this study.

Rating Scores by Case Specificity

Regarding the levels of specificity in case profiles within prompts (e.g., *Basic, Intermediate,* and *Detailed* case descriptions), the group average scores of intervention plans generated from prompts with Detailed case descriptions were slightly higher than those from prompts with Basic case descriptions. The outputs generated from Simple Commands exhibited this pattern. The group average scores of outputs from Framed Commands, the outputs from Framed Commands with Detailed and Basic case descriptions received slightly higher scores than those with Intermediate case descriptions. For example, the average scores of the outputs for LD cases were 1.90 (SD = 0.51), 1.93 (SD = 0.55), and 2.02 (SD = 0.49) for Simple Commands with Basic, Intermediate, and Detailed case

Table 2. Summary of average ratings and standard deviations of artificial intelligence (AI) outputs.

		ASD								LD			
		Simple Commands				Framed Commands				Simple Commands			
AI tools	Rating scores	Basic	Inter	Detailed	Ave	Basic	Inter	Detailed	Ave	Basic	Inter	Detailed	Ave
ChatGPT	Average scores	2	2.07	2.27	2.11	2.57	3.17	3.27	3.00	2.1	1.83	2.3	2.08
	Standard deviations	0.65	0.7	0.76	0.70	0.5	0.85	0.98	0.78	0.58	0.56	0.57	0.57
ChatGPT-4o	Average scores	2.67	2.43	2.33	2.48	3	2.93	3.13	3.02	2.43	2.2	2.37	2.33
	Standard deviations	0.63	0.56	0.62	0.60	0.6	0.64	0.81	0.68	0.5	0.39	0.57	0.49
Copilot	Average scores	1.43	1.43	1.43	1.43	2.4	2.13	2.23	2.25	1.47	1.77	1.47	1.57
	Standard deviations	0.5	0.4	0.34	0.67	0.5	0.43	0.46	0.46	0.41	0.59	0.33	0.44
Gemini	Average scores	1.93	2.2	2	1.91	3.17	3.33	2.87	3.12	2	2.07	2.3	2.12
	Standard deviations	0.54	0.77	0.7	0.67	0.51	0.84	0.6	0.65	0.53	0.66	0.61	0.60
KiKi Creates	Average scores	1.37	1.43	1.6	1.47	2.57	2.43	2.8	2.60	1.5	1.5	1.57	1.52
	Standard deviations	0.42	0.5	0.31	0.41	0.58	0.5	0.69	0.59	0.33	0.56	0.41	0.43
MagicSchool	Average scores	1.87	2.13	2.07	2.02	2.5	1.93	2.43	2.29	1.87	2.2	2.13	2.07
	Standard deviations	0.61	0.62	0.46	0.56	0.58	0.54	0.57	0.56	0.7	0.52	0.42	0.55
Total	Average scores	1.81	1.95	1.95	1.98	2.70	2.65	2.79	2.71	1.90	1.93	2.02	1.95
	Standard deviations	0.56	0.59	0.53	0.60	0.55	0.63	0.69	0.62	0.51	0.55	0.49	0.51

	LD				SSD								Overall average					
	Framed Commands				Simple Commands				Framed Commands				ASD	LD	SSD	Simple	Framed	Overall
AI tools	Basic	Inter	Detailed	Ave	v	1.2	1.3	Ave	Basic	Inter	Detailed	Ave						
ChatGPT	2.87	2.67	2.77	2.77	1.7	2.13	1.9	1.91	2.7	2.57	2.83	2.70	2.56	2.42	2.31	2.03	2.82	2.43
	0.46	0.51	0.64	0.54	0.42	0.5	0.36	0.43	0.41	0.5	0.57	0.49	0.74	0.55	0.46	0.57	0.60	0.58
ChatGPT-4o	3.17	3.17	3.1	3.15	2.43	2.43	2.37	2.41	3.3	2.83	2.93	3.02	2.75	2.74	2.72	2.41	3.06	2.73
	0.48	0.62	0.67	0.59	0.64	0.6	0.7	0.65	0.64	0.63	0.7	0.66	0.64	0.54	0.65	0.58	0.64	0.61
Copilot	2.2	2.3	2.53	2.34	1.7	1.6	1.67	1.66	2.27	2.4	2.37	2.35	1.84	1.96	2.00	1.55	2.31	1.93
	0.38	0.62	0.5	0.50	0.47	0.62	0.47	0.52	0.47	0.56	0.52	0.52	0.57	0.47	0.52	0.54	0.49	0.52
Gemini	2.83	2.93	2.7	2.82	1.77	1.83	1.9	1.83	2.97	2.77	3.17	2.97	2.58	2.47	2.40	2.00	2.97	2.49
	0.48	0.51	0.51	0.50	0.41	0.45	0.64	0.5	0.57	0.61	0.65	0.61	0.66	0.55	0.56	0.59	0.59	0.59
KiKi Creates	2.37	2.63	3	2.67	1.57	1.53	1.87	1.66	2.43	2.47	2.83	2.58	2.03	2.10	2.12	1.55	2.61	2.08
	0.48	0.45	0.57	0.50	0.42	0.48	0.63	0.51	0.56	0.62	0.48	0.55	0.50	0.47	0.53	0.45	0.55	0.50
MagicSchool	2.2	2.33	2.17	2.23	1.87	2.03	2.3	2.07	1.91	2.2	2.03	2.05	2.16	2.15	2.06	2.05	2.19	2.12
	0.46	0.48	0.31	0.42	0.46	0.44	0.51	0.47	0.51	0.62	0.55	0.56	0.56	0.48	0.52	0.53	0.51	0.52
Total	2.61	2.67	2.71	2.66	1.84	1.93	2.00	1.92	2.60	2.54	2.69	2.61	2.32	2.31	2.27	1.93	2.66	2.30
	0.46	0.53	0.53	0.51	0.47	0.52	0.55	0.51	0.53	0.59	0.58	0.57	0.61	0.51	0.54	0.54	0.56	0.55

Note. ASD = autism spectrum disorder; LD = language disorder; SSD = speech sound disorder; Inter = Intermediate; Ave = average.

descriptions, respectively. The outputs generated from Framed Commands with Basic, Intermediate, and Detailed case descriptions received average scores of 2.61 (SD = 0.46), 2.67 (SD = 0.53), and 2.71 (SD = 0.53), respectively. Similar results were observed in Simple Command prompts in SSD and ASD cases. The differences in the group average values were small, making it difficult to claim that they were meaningful differences. However, a similar pattern appeared across the three disorder cases. These results may indicate that, when prompts included Simple Command statements, detailed case descriptions may generate relevant intervention plans compared to the prompts with less information. On the other hand, when prompts included Framed Command statements, the specificity of the case information may not improve output performance. Mostly, Framed Command prompts paired with either basic or highly detailed case information yielded the best performance, whereas intermediate case descriptions resulted in lower performance.

When investigating the average ratings across the specificity levels for each AI tool, no consistent patterns were observed. Some AI tools produced more accurate and relevant intervention plans as the case descriptions included more detailed information. For example, ChatGPT-3.5, KiKi Creates, and MagicSchool received a higher rating for outputs from Simple and Framed Commands with Detailed case descriptions compared to those of prompts with Basic case descriptions. However, ChatGPT-4o's output deteriorated as case specificities were increased for both Simple and Framed Commands. Gemini produced the best outputs for Simple and Frame Commands with Intermediate case descriptions. Copilot produced inconsistent results, which made it difficult to find general patterns in this tool.

Highest Scoring Items

This study analyzed the highest scoring items in the grading rubrics for each AI tool, highlighting the strengths of the intervention plans generated by each AI tool in specific areas. The grading rubric included six objective items to evaluate: (#1) therapy goals, (#2) evidence-based intervention strategies, (#3) therapy activities, (#4) client engagement, (#5) progress monitoring tools, and (#6) cultural, linguistic, and family considerations. In the analysis, the items that received the highest average scores for each AI tool were identified by their item numbers, indicating which of the six areas were addressed most relevantly and accurately by the AI tool.

The highest-scoring items for ChatGPT-3.5 were #3 and #4, indicating therapy activities and client engagement were the strong areas of intervention plans generated by the tool. The highest-scoring item for ChatGPT-4o was #1, which involved detailed and relevant therapy goals. Gemini produced various therapy activities (#3) and cultural and family considerations (#6). Copilot, KiKi Creates, and MagicSchool included various ways to encourage client engagement (#4). In all AI tools, #2 and #5 were rarely scored high, indicating that evidence-based practice and progress monitoring may be the areas to be improved for these AI tools. The results can be found in Table 3.

Qualitative Evaluations

The qualitative analysis of the researchers' responses to an open-ended question revealed common themes regarding the strengths and limitations of these AI tools' outputs. The common themes included the inclusion of required elements, a structured format, specificity of information, transferability to real cases, client-specific considerations, research evidence for intervention strategies, progress monitoring tools, client engagement strategies, and cultural and family considerations.

The analysis identified several strengths across the AI tools. Most AI tools' outputs such as ChatGPT-3.5, ChatGPT-4o, Gemini, and KiKi Creates included required fundamental elements in a well-structured format. Although Copilot and MagicSchool were missing some components, they still delivered reports in a well-structured format. Most AI tools offered a range of therapy activities and strategies for client or family engagement. These tools showed the improvement in generating goals and intervention strategies with prompts with Detailed case descriptions. This was particularly evident in ChatGPT-4o outputs, where researchers noted, "Improved goals and intervention strategies, and activities than ChatGPT 3.5." ChatGPT-4o and Gemini showed improvement in cultural and family considerations, with "cultural considerations" appearing more frequently in their detailed case description outputs. However, other tools such as Kiki Creates and Copilot often had "missing cultural considerations" noted as an area for improvement.

Common limitations across all AI tools included a lack of specificity and transferability. Most intervention plan outputs were described as too general and lacking client-specific details. Another finding across all AI tools was the consistent weakness in generating progress monitoring tools and evidence-based intervention strategies supported by research. This was evident in frequent comments such as "lacking progress monitoring tool" (ChatGPT SSD, Framed with Intermediate), "lacking EBP and progress monitoring" (Gemini ASD, Framed with Detailed), and "lacks EBP" (Copilot, multiple instances). Even for better-performing tools such as ChatGPT-4o, areas for improvement often included "progress monitoring tools and cultural considerations."

Table 3. The highest scored item numbers of artificial intelligence (AI) outputs (#1: therapy goals, #2: evidence-based intervention strategies, #3: therapy activities, #4: client engagement, #5: progress monitoring tools, and #6: cultural, linguistic, and family considerations).

	ASD						LD						SSD					
	Simple Command			Framed Command			Simple Command			Framed Command			Simple Command			Framed Command		
AI tools	Basic	Inter	Detailed	Basic	Inter	Detailed	Basic	Inter	Detailed	Basic	Inter	Detailed	Basic	Inter	Detailed	Basic	Inter	Detailed
ChatGPT-3.5	3	1	3	3, 4	4	4	3	3	3	3, 4	4, 6	4	3	1	3	4	4, 6	6
ChatGPT-4o	3	1	1	2	2, 3, 4, 6	6	1	1	1	1	4	6	1, 2	1	1	6	4	2
Copilot	6	3, 5	4	1	4	2, 4	3	3	3	1	6	1	3, 6	3, 6	6	4	2, 4	4
Gemini	3	1	6	1, 2, 3, 4	1, 3, 4	4	3	3	1	1	3, 4	3, 6	6	6	6	6	3, 4	6
KiKi Creates	1, 3, 4	3	3, 4	1	1	1, 2, 3, 6	4	3	3	3	1, 4	1	6	4, 6	6	6	4	2, 4, 6
MagicSchool	4	1	4	4	4	4	1	1	4	1	1, 4	1	3	1	1	4	4	1, 4

Note. ASD = autism spectrum disorder; LD = language disorder; SSD = speech sound disorder; Inter = Intermediate.

The inconsistent impact of prompt specificity on output quality was also mentioned in the qualitative analysis. Prompts with Detailed prompts often led to more detailed and relevant outputs, but this improvement was not uniform across all tools or components. For instance, with ChatGPT-4o, researchers noted that some prompts with detailed case descriptions produced outputs "similar to, or worse than, the lower level" (ASD, Framed with Intermediate), highlighting that increased prompt specificity did not always yield better results. For many AI tools, the more detailed prompts such as Framed with Intermediate prompts frequently resulted in lower-quality performance than Framed with Basic prompts.

The qualitative analysis also identified overall characteristics unique to each AI tool. ChatGPT-3.5 produced intervention plans, which include relevant information in a well-structured format, but outputs were too general, included too many related items without prioritizing, and lacked specificity and transferability. Researchers frequently noted "decreased quality" for detailed prompts (e.g., SSD, Simple with Detailed) and "lacks relevance" (SSD, Framed with Detailed). The qualitative data for ChatGPT-4o suggested that its output characteristics were similar to that of ChatGPT-3.5 but showed improvement over ChatGPT-3.5, especially goals and therapy activities. ChatGPT-4o was more consistent in addressing relevant intervention strategies, research references, and cultural considerations across cases and prompt levels. However, its outputs remained too general, lacking both transferability and client-specific details.

Gemini demonstrated improvements in clinical relevance and client-specific strategies as prompt levels increased. Researchers observed "improvement from prompts with Basic to prompts with Intermediate case descriptions" and "prompts with Detailed case descriptions generated more detailed, client-relevant plans" across cases, indicating its potential for generating more tailored intervention plans with more specific inputs. However, relevant goals, activities, progress monitoring tools, and client-specific considerations were lacking, which limited the transferability of the outputs to real cases.

For Kiki Creates, Copilot, and MagicSchool, deficiencies were consistently noted in most areas, including goals, intervention strategies, research evidence for intervention strategies, and progress monitoring tools. For Kiki Creates, comments such as "lacking EBP and progress monitoring" were common across cases. Copilot was frequently described as "not transferable" and "too general, not specific to client" across all prompt levels and cases. MagicSchool included various engagement strategies and family involvement but lacked relevant speech and language goals as well as effective intervention strategies. Quantitative results summarizing the key characteristics, strengths, and limitations of each tool are summarized in Table 4.

Discussion

Overall Evaluation of Intervention Plan Outputs

This study analyzed the overall average rating scores for all intervention plan outputs generated by six currently available AI tools. The overall average score from all six AI tools was 2.31 (SD = 0.55), ranging from 2.01 to 2.73, indicating that the intervention plan outputs generated by these AI tools were rated at a level between 3 (*Meet Expectations*) and 2 (*Needs Improvement*). When considering the average scores for each tool, ChatGPT-4o, Gemini, and ChatGPT received slightly higher scores than MagicSchool, KiKi Creates, and Copilot, in that order. It should be noted that even ChatGPT-4o, which received the highest average score of 2.73 (SD = 0.61), did not achieve a rating of 3.

The ratings in this study's grading rubric were designed to be comparable to the clinical knowledge and competency required for speech-language pathology clinicians. A rating of 3 (*Meets Expectations*) was equivalent to a C grade when compared to the learning outcomes of human graduate students in speech-language pathology. The results of this study revealed that, despite variability in the average scores across these six AI tools, their intervention plan output did not *meet* or *exceed* the minimum expectations for clinical knowledge and competency. This finding aligns with other recent research findings, which report that human-written clinical writing tends to be more concrete and diverse, and typically contains more useful information compared to AI-generated outputs (e.g., Liao et al., 2023). On the other hand, all these six tools rated higher than a rating of *Unsatisfactory*, indicating that these tools may have the potential to become useful resources for clinical writing in speech-language pathology with further improvement.

Impacts of Prompts

The rating scores of outputs generated by each AI tool varied depending on the characteristics of the prompts. In this study, two types of command statements, Simple Command and Framed Command, were used to generate outputs. All six AI tools produced outputs with higher ratings when used with prompts with Framed Commands compared to prompts with Simple Commands. Most notably, ChatGPT-4o consistently produced

Table 4. Summary of qualitative analysis across six artificial intelligence (AI) tools.

AI tool	Strengths	Weaknesses	Prompt level differences	Overall characteristics
ChatGPT-3.5	- Comprehensive inclusion of plan elements - Structured format - Goals and therapy activities - Improved engagement strategies at higher levels - Some cultural considerations	- Lack of specificity and transferability - Not client specific - Weak evidence-based practice (EBP) - Weak progress monitoring tool - Inconsistent quality across cases	- Framed Command prompts showed a slight improvement in engagement and intervention strategies - Quality sometimes decreased in Intermediate or Detailed prompts	- Demonstrates potential but lacks consistency - Too general - Higher specificity prompts do not always yield better results
ChatGPT-4o	- Comprehensive inclusion of plan elements - Structured format - Improved goals, intervention strategies, and activities than ChatGPT-3.5 - Included cultural considerations and family involvement	- Lack of specificity and transferability - Not client specific - Inconsistent quality across prompt levels - Progress monitoring tool often missing or lacking	- Framed Command prompts showed improved transferability - Some redundancy in higher-level prompts - Quality sometimes decreased in Intermediate prompts or three prompts	- Shows improvement over the 3.5 version, especially goals and activities - Still too general - More consistent in addressing cultural aspects - Struggles with maintaining quality across all prompt levels
Copilot	- Structured format - Occasional EBP citations	- Too general and there are many missing components - Lack of specificity and transferability - Not client specific - Goals, EBP, and progress monitoring tools are lacking	- Slight improvement in structure with higher-level prompts - Inconsistent quality across levels	- Too general - Significant struggles with specificity and relevance - More specific case description prompts do not consistently improve output quality - Not relevant for clinical writing
Gemini	- Comprehensive inclusion of plan elements - Structured format - Improved clinical relevance at higher levels, especially for goals, EBP, and activities - Occasional client-relevant goals - Occasional relevant EBP citations - Cultural and family considerations	- Lack of specificity and transferability - Not client specific - Inconsistent inclusion of components - Weak progress monitoring	- Framed Command prompts showed more detailed, client-relevant plans than Simple Command prompts	- Lack of specificity and transferability - Too general - Shows potential for improvement with more specific prompts - Struggles with consistent inclusion of all necessary components in Detailed prompts compared to Basic and Intermediate prompts

(*table continues*)

Table 4. *(Continued).*

AI tool	Strengths	Weaknesses	Prompt level differences	Overall characteristics
KiKi Creates	- Some clinically relevant and specific goals - Incorporates specialized terminology relevant to speech-language pathology - Relevant activities - Improved intervention strategies at higher levels	- Lack of specificity and transferability - Lack of client-specific considerations - Lacking EBP and progress monitoring - Missing cultural considerations	- Gradual improvement from Basic to Detailed prompts	- Use relevant terminology but still too broad - Shows improvement with higher-level prompts - Lack of specificity and transferability - Consistently struggles with EBP and progress monitoring tools
MagicSchool	- Suggestions for client-relevant motivation or family engagement strategies - Occasional cultural considerations	- Lack of specificity and transferability - Missing components - Goals, intervention strategies - Weak EBP and progress monitoring	- Improvement from Basic to Intermediate prompts, but not Detailed prompts	- Too general - Not relevant for clinical purposes directly, but may be used when interprofessional collaboration in a school setting

intervention plans with ratings higher than *Meet Expectations* for all three disorders when prompts with Framed Commands were entered. In contrast, the average scores of outputs from Gemini and ChatGPT-3.5 reached the *Meets Expectation* level for ASD only. No intervention plans from Copilot, KiKi Creates, and MagicSchool met expectations for any prompts. These results indicate that command prompts requiring specific output format structures may yield more targeted and relevant intervention plan outputs compared to general prompts, even though the consistency in meeting clinical standards may vary among different AI tools.

Unlike the types of prompts, the level of specificity in case information within prompts did not consistently impact the quality of outputs in this study. Each prompt consisted of one of two types of command statements, followed by a case description containing three levels of specific case descriptions (*Basic*, *Intermediate*, and *Detailed*). Since these AI tools generated outputs based on inputs as well as online sources, it was expected that more detailed input might yield more relevant outputs. However, consistent output patterns were not observed across different levels of prompt specificity.

The researchers observed that Simple Command prompts produced more accurate and relevant intervention plans as the following case descriptions included more detailed information, especially when comparing prompts with Simple Commands with Basic and Detailed case descriptions. However, outputs from Simple Commands with Intermediate case descriptions received lower, higher, or sometimes the same ratings as those from Simple Commands with Basic case descriptions. In Framed Command prompts, the variability in output quality based on the specificity of case information was more prominent. ChatGPT-4o's and ChatGPT-3.5's outputs deteriorated as case specificities increased from Framed Commands with Basic case descriptions to Framed Commands with Intermediate case descriptions. The outputs from Framed Commands with Detailed case descriptions, generally showed improvement, but in some cases, they deteriorated further compared to the output from Framed Commands with Intermediate case descriptions. Gemini and MagicSchool produced the best outputs for Framed with Intermediate prompts compared to Framed with Basic and Framed and Detailed prompts. Other AI tools showed highly inconsistent results across the three levels.

These findings do not fully support the suggestions of recent articles (e.g., Price et al., 2024), which propose that providing as much information as possible in the prompt leads to better performance. Rather, the authors of this study suggest that when prompts consist of command statements without specific output format requirements, providing detailed case information may produce more accurate and relevant intervention plans. In contrast, prompts containing command statements with specific format requirements may yield relevant intervention plans compared to prompts with Simple Command statements, but the level of detail in the case information may have little effect or, in some cases, negatively impact the quality of the intervention plan outputs. For most AI tools analyzed in this study, specific command prompts paired with either basic or detailed case information yielded slightly higher performance, whereas intermediate level of case descriptions resulted in lower performance.

It is unclear whether these results are related to the specificity levels of case information, the interactions between prompt types and specificity levels, or the repetition of similar input entries to these AI models. It was noticed that the performance of these AI tools deteriorated after a certain number of consecutive similar input prompts entry. These reliability issues should be investigated in future research.

Clinical Potential, Strengths, and Limitations of AI Tools

Each AI tool investigated in this study possesses unique characteristics, strengths, and weaknesses in supporting clinical writing. ChatGPT-3.5 and ChatGPT-4o, which share similar algorithms but differ in improved performance, generate diverse and detailed information in their intervention plans compared to other AI tools analyzed in this study. ChatGPT-4o produced intervention plans with improved goals and slightly more detailed information than ChatGPT-3.5, although the difference was not significant. On the other hand, these tools often overload with too many related items without prioritizing, resulting in plans that lack transferability and client specificity. Overall, ChatGPT-3.5's and ChatGPT-4o's intervention plans appeared comparable to that of a new student clinician who understands the basics but cannot yet select the most relevant items for a case.

Gemini, Google's LLM AI, utilizes the Google search engine to produce results, benefiting from constantly updated online information. Even though Gemini's outputs included fewer items and did not adhere to clinical writing formats, it provided useful information in therapy goals, intervention strategies, and activities, particularly with detailed and structured command prompts. However, Gemini's outputs still remained too general and lacked transferability. The authors suggest that integrating the strengths of both Gemini and ChatGPT series could result in more effective outcomes.

In contrast, Microsoft Copilot is an AI tool specifically designed to support business tasks using Microsoft software. Due to its nature, the outputs generated by Copilot were not suitable for clinical documentation formats and content. Although outputs from specific command prompts received higher ratings than those from common prompts, the information provided was limited and repetitive across different cases.

KiKi Creates produced intervention plan outputs with relevant contents and effectively used professional terminology. With detailed and structured prompts, the outputs contained relatively relevant therapy goals and intervention strategies. However, the outputs remained too broad, not client specific, and lacked transferability. Additionally, research evidence to support the intervention strategies was absent. Since this tool uses an internal database rather than open sources, its suggested activities and materials were limited and repetitive, regardless of the case or client age.

Lastly, MagicSchool demonstrated its usefulness as a tool for generating ideas to support students with speech and language deficits in academic settings. However, the intervention plans were overly general, presumably due to predetermined structures designed to support classroom teaching for educators. This tool may be useful for providing supplementary support to clinicians when collaborating with classroom teachers.

In summary, most AI tools analyzed in this study generated intervention plan outputs in a well-structured format, offering various activities and general client engagement strategies. However, they lacked relevant therapy goals, research-evidenced intervention strategies, and cultural and linguistic considerations. Specifically, general-purpose AI tools utilizing LLMs, such as ChatGPT-3.5, ChatGPT-4o, and Gemini produced intervention plans that included a wider variety of activities, materials, and cultural considerations compared to the more focused AI products, Copilot, MagicSchool, and KiKi Creates. None of the AI tools sufficiently addressed individual needs or provided client-specific progress monitoring tools. These findings indicate that while these tools have potential as general guidelines for speech-language pathology intervention plan writing, they are not yet fully capable of producing transferable and individualized plans. Our findings align with recent research (e.g., Lee et al., 2024; Zhong, 2024) suggesting that future directions in AI-assisted clinical documentation should focus on developing more reliable AI models, conducting rigorous empirical research to validate AI applications, and establishing clear legal and ethical guidelines for AI use in health care.

Limitations of This Study and Future Directions

This study has several limitations. The researchers analyzed the outputs from only a single attempt per model for each prompt. As an early-stage investigation, this study focused on identifying broad patterns across prompts. However, conducting multiple attempts per model would enhance the robustness of comparison results. We will further explore the impact of multiple attempts on output quality in future research. In this study, we aimed to explore prompts that typical application users without computer science backgrounds might naturally use in a zero-shot setting. We believe this approach provides valuable insights into real-world interactions with LLMs. However, we acknowledge that there are various well-established prompting strategies. These considerations should be included in future research. The researchers observed a deterioration in performance after a certain number of continuous attempts with different prompts for the same case. Future research needs to systematically investigate how the interactions or collaborations with AI tools influence the intervention plan outputs.

This study compared AI performance using the average scores provided by a small number of raters. Future research would benefit from increasing the number of raters and incorporating statistical significance tests to assess meaningful differences and provide more robust and reliable findings. Moreover, researchers analyzed pediatric cases with ASD, LD, and SSD. Future studies should evaluate intervention plans for cases with other disorders or different age populations. Lastly, future research should explore specialized LLMs to expand the findings. Given the rapid advancement of AI technologies, ongoing research is essential to keep the research community updated. As part of future work, we plan to develop a specialized LLM by fine-tuning existing models with speech-language pathology–related data and conducting a comprehensive evaluation.

Conclusions

This study aimed to evaluate the application of currently available AI tools in supporting clinical report writing within clinical and educational settings. While these AI tools show potential for providing general guidelines for writing speech-language pathology intervention plans, they are not capable of producing transferable and clinically relevant plans that address individual needs. As AI technology continues to develop, it is crucial for clinicians, educators, and students in communication sciences and disorders and related areas to understand the unique characteristics, strengths, and limitations of these AI tools to use them effectively and appropriately. The researchers believe that this study's outcomes may positively contribute valuable insights to inform best clinical practices for AI use moving forward. It is essential that ongoing discourse and continual examination of ethical and professionally responsible use of ever-evolving AI technologies for speech-language pathology and related professions remains in play.

Data Availability Statement

All data generated or analyzed during this study are included in this published article. However, the further details of data sets generated and/or analyzed during the current study are also available from the corresponding author upon reasonable request.

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Appendix A

A Sample SSD Case Profile With Three Specificity Levels

Basic Case Description:

The client, David, is a 5-year-old Spanish–English bilingual male with phonological-based speech sound disorders. He demonstrated a limited consonant inventory, speech error patterns, and poor intelligibility.

Intermediate Case Description:

The client, David, is a 5-year-old Spanish–English bilingual male with phonological-based speech sound disorders. He demonstrates language proficiency in both Spanish and English, using English in the school setting. He demonstrated limited consonant inventory, limited syllable structures, several speech error patterns, and poor intelligibility. His inventory includes bilabial and alveolar stops, nasals, and glides, but not other consonants. His speech errors include final consonant deletion, fronting, stopping, gliding, and cluster reduction.

Detailed Case Description:

The client, David, is a 5-year-old Spanish–English male with phonological-based speech sound disorders. He presents with a limited consonant inventory, limited syllable structures, several speech error patterns, and poor intelligibility. He produces bilabial and alveolar stops, nasals, and glides, but not other consonants. His speech errors include final consonant deletion, fronting, stopping, gliding, and cluster reduction. He was referred by his pre-K teacher and the teacher reported that she can understand him less than 40% of the time. David's parents reported that they can understand most of the time, but other family members and friends have difficulty understanding him.

David's family is originally from Puerto Rico. He demonstrates language proficiency in both Spanish and English. English is used in the school setting, while both Spanish and English are used at home. Per his parents' report, David was born healthy without any complications and had no serious illnesses or injuries since then. He reached typical developmental milestones, but his first words only appeared around age 2. David lives with his parents and his younger brother who is 8 months old and often spends time with his grandparents and other relatives. The family speaks in English only. There is no history of speech or language deficits in the family. He is physically healthy and remains active daily. His teacher reported that David is academically sound and very good at following directions in class. His physical, cognitive, and hearing abilities are within developmental age expectations. No concerns have been reported regarding his social abilities.

Assessment Results:

The structure and function (e.g., tone, mobility, and range of motion) of the speech mechanism are adequate for speech production. According to GFTA-3 standardized test measures, David yielded a raw score of 42, a standard score of 62, and registered in the 3rd percentile rank. These scores indicate David is performing below average for a child of David's age and can infer that, out of 100 children, 97 of them would score higher in producing consonant sounds.

According to spontaneous speech sample analysis, David demonstrated his strengths in early-developing consonants. He produced stop sounds (excluding velar stop sounds /k/ and /g/), nasal sounds (excluding the velar /ŋ/), and glides in initial and medial word positions. However, he did not produce velar consonants and other late-developing consonants such as fricatives, affricates, and liquids in all word positions. David was stimulable for velar consonants /k/, /g/, and the fricatives /f/ and /v/ in initial and medial word positions. According to phonological process analysis, David produced the following phonological processes: final consonant deletion, velar fronting (ex. /d/ for /g/ or /t/ for /k/), stopping of fricatives and affricates (ex. /t/ for /s/ or /d/ for /z/), liquid gliding /w/ for /l, r/, and cluster reduction (ex. /bw/ for /br/ or /d/ for /dr/).

His receptive and expressive language abilities are within age expectations. However, he does not use age-expected grammatical morphemes including plurals, verb endings, and present progressive *–ing* due to his limited consonant inventory and syllable structures.

Appendix B

Grading Rubrics

Quantitative items

#1. Goal recommendations based on case history and assessment provided
- 1: Goal recommendations not provided
- 2: Some goal recommendations were provided but broad
- 3: Goal recommendations are somewhat client-relevant but somewhat general
- 4: Goal recommendations are generally client-relevant with some client specificity
- 5: Goal recommendations are highly relevant and specific to the client's individual needs.

#2. Evidence-based intervention strategies
- 1: No implicitly cited evidence-based practices for intervention
- 2: Some cited evidence-based interventions, but not specific to the client
- 3: Evidence-based interventions recommended are general and somewhat client-relevant
- 4: Evidence-based interventions recommended are generally client-relevant with some client specificity
- 5: Evidence-based interventions recommended are highly relevant and specific to the client's individual needs (with 2 research references)

#3. Age-appropriate therapy activities and materials
- 1: Age-appropriate recommendations for therapy activities and materials not provided
- 2: Some age-appropriate therapy activities and materials were provided but not specific to the client
- 3: Age-appropriate therapy activities and materials provided are general and somewhat client-relevant
- 4: Age-appropriate recommendations for therapy activities and materials provided are generally client-relevant with some client-specificity
- 5: Age-appropriate recommendations for therapy activities and materials provided are highly relevant and specific to the client's individual needs.

#4. Client engagement, behavior and motivation
- 1: No reference to client engagement, behavior, and motivation strategies
- 2: Minimal verbiage regarding client engagement, behavior, and motivation strategies
- 3: Some general client engagement strategies suggested, not specific to the client's needs
- 4: A few very client-specific engagement, behavior, and motivation strategies suggested
- 5: Several suitable client-specific engagement, behavior, and motivation strategies suggested

#5. Progress monitoring tools for tracking goal progress
- 1: No progress monitoring tools provided
- 2: Progress monitoring tools provided but broad and not specific to client goal tracking
- 3: Progress monitoring tools suggested are very general for goal tracking
- 4: Progress monitoring tools suggested are generally relevant for tracking client goals
- 5: Progress monitoring tools suggested are highly relevant and specific for tracking client goals

#6. Culturally responsive and inclusive practice/sensitivity
- 1: Lack of consideration for cultural background in intervention planning
- 2: Minimal attention to cultural sensitivity in interventions
- 3: General cultural considerations stated for cultural diversity in intervention
- 4: Some specific cultural considerations suggested for intervention
- 5: Several specific culturally inclusive and responsive suggestions/considerations made, tailored to the client's background

Qualitative item

#1. How would you describe the quality of the intervention plan outputs generated by the tool? Additionally, if possible, please identify 1–3 strengths and areas for improvement for each output.